Nihar Kalode

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Education

University of Illinois Urbana-Champaign Urbana, IL
B.S. in Mathematics & Computer Science, BSLAS in Statistics — GPA: 3.98 *Aug 2024 – May 2028*

- **Coursework:** Deep Generative Models, Computer Vision, Machine Learning, Statistical Reinforcement Learning, Probability Theory, Real Analysis, Measure-Theoretic Analysis, Mathematical Statistics, Abstract Linear Algebra

Publications

Sparse Similarity-Aware Label Smoothing via Learned Latent Spaces **[PDF] [Code]**
Nihar Kalode — [Under Review, NeurIPS 2026]

Experience

Student Researcher May 2026 – Present
Advised by Prof. Svetlana Lazebnik @ UIUC | PyTorch, Diffusers, Diffusion Models

- Leading independent research toward a first-author publication at the intersection of diffusion models and representation learning, particularly in the image generation domain, working one-on-one with Professor Lazebnik.
- Audited the Universal Normal Embedding framework's claim of a shared Gaussian embedding space underlying representation learners and diffusion model noise, testing universality, normality, and linear-probe generalization.
- Showed embeddings have much weaker Gaussian structure than reported using proper high-dimensional statistical testing, semantic edits attributed to noise inversion were reproducible via direct VAE latent editing, and linear-probe trends failed to generalize to more challenging settings, disproving the framework's central claims.
- Redirecting the project toward new research directions in diffusion models, continuing into the fall semester.

Summer Research Intern May 2025 – Aug 2025
Parasol Lab @ UIUC | C++, CUDA, MPI

- Integrated multi-GPU execution into open-source C++ STAPL Library using the CUDA Runtime API and Thrust; achieved up to 100x speedups and improving computational efficiency for large-scale data processing.
- Developed robust CMake-based build automation and cross-platform testing pipelines across 1-16 distributed MPI systems, enhancing reliability and transparency in parallel algorithm performance.

Projects

Semantic Region-Aware Diffusion | *PyTorch, Diffusion Controllability, LoRA* March 2026 – May 2026

- Proposed a lightweight extension of Region-Aware Diffusion incorporating segmentation-map guidance via multi-scale feature injections throughout the U-Net, enabling semantic controllability with minimal overhead.
- Ran qualitative editing experiments directly modifying the conditioning segmentation map, verifying that edits produced corresponding structural changes in the generated image output.
- Quantified conditioning strength via feature-perturbation analysis across U-Net layers and timesteps, benchmarking injection strategies and finding downsampling injection outperformed full-network injection (FID 31.1 vs. 32.2).

High-Performance Memory Allocator | *C, Systems Programming, Memory Management* Feb 2026

- Designed and implemented a custom dynamic memory allocator with a novel segregated free-list structure, using it as a testbed to study fragmentation and allocation efficiency trade-offs.
- Investigated adaptive splitting and coalescing policies, empirically characterizing how threshold and minimum-block-size parameters affect internal vs. external fragmentation across varied allocation patterns.
- Benchmarked allocation and search efficiency across diverse workloads, achieving a score of 122.96 and outperforming the optimized `glibc` allocator baseline (∼115), validating the proposed adaptive strategy.

Technical Skills

Languages: Python, C, C++, R, LaTeX
ML/Research Tools: PyTorch, Diffusers, Hugging Face, NumPy, SciPy, Pandas, Scikit-learn, Matplotlib
Research Areas: Diffusion Models, Representation Learning, Neural Network Calibration, Conditional Generation
Systems & Infra: CUDA, MPI, CMake, Docker, Git